

```
# Graph represented as an adjacency list
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F', 'G'],  
    'D': [],  
    'E': [],  
    'F': [],  
    'G': []  
}
```

```
def dfs(graph, node, visited):
```

```
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)
```

```
    # Visit all adjacent nodes recursively  
    for neighbor in graph[node]:  
        dfs(graph, neighbor, visited)
```

```
# Driver code
```

```
visited = set()  
print("DFS Traversal:")  
dfs(graph, 'A', visited)
```

```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct 4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hi/OneDrive/Desktop/AI/expp.py =====
DFS Traversal:
A B D E C F G
>>>
```

```
# Graph represented as an adjacency list
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        # Visit all adjacent nodes recursively
        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)

# Driver code
visited = set()
print("DFS Traversal:")
dfs(graph, 'A', visited)
```